

PADMAPRIYA RAJASEKARAN

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SUMMARY

Biomedical Engineering graduate with 1.5 years of hands-on experience executing system-level Verification & Validation (V&V), integration testing, and root cause analysis on FDA-regulated electromechanical devices

EDUCATION

Rochester Institute of Technology

B.S., Biomedical Engineering | Chemistry Minor

Aug 2021 – Dec 2026

Dean's List

SKILLS

V&V, RCA, IQ/OQ/PQ Concepts, Traceability Matrices (RTM), Risk-Based Validation, Gage R&R, TP Authoring, MATLAB, FEA, WI & SOP Authoring, Functional & Regression Testing, Deviation & Non-Conformance Management, Change Control, GDP, cGMP Experience

WORK EXPERIENCE

Fresenius Medical Care

Systems Engineering Co-op, NPI

Jul 2025 – Dec 2025

Lawrence, MA

- Contributed to resolving a known temperature-sensitivity issue behind recurring balance-test failures by rebuilding the test fixture with in-line sensors and a calibrated scale, **restoring test validity.**
- Confirmed test station and fixture installations met spec (IQ) and executed OQ/PQ protocols verifying system operation and performance, **clearing systems for use without rework.**
- Executed V&V protocols (balance, temperature, leak) across 20+ system configurations, generating traceable datasets supporting formal FDA verification submissions.
- Ran system-of-systems integration testing across 5 test stations interfacing development hardware with commercial platforms, confirming consistent performance before release to validation.
- Tested temperature and flow-rate parameter combinations during root cause analysis, **identifying values that met performance targets and were adopted into the final test protocol.**
- Analyzed 100,000+ data points per test run in MATLAB to surface pressure-weight error correlations, flagging anomalies the team used to refine test parameters.
- Troubleshooted a closed-loop temperature controller's software-hardware interface, tracing inconsistent readings to a thermocouple/setpoint mismatch.
- Continuously monitored real-time system behavior (valve states, internal flow) via a SCADA-style interface, **building a reference baseline that accelerated later troubleshooting.**
- Authored MATLAB analysis workflows for pressure and weight sensor data, **later adopted by other engineers on the team.**
- Verified alarm, GUI, and fault-handling logic across 7 systems via fault-injection testing, **confirming safe system shutoff under simulated failure conditions.**
- Executed conductivity sensor verification across a 5-variable matrix and applied SPC across 20+ trials, **confirming sensor accuracy and repeatability within specification.**

Cranial Technologies

R&D Engineering Co-op

Jul 2023 – May 2024

Chandler, AZ

- Executed verification testing of MATLAB-based shell-generation outputs across 4 band types using specialized 3D inspection software, **identifying defects across 100+ cases per type and routing failures back for correction.**
- Inspected STL files for geometry defects prior to nesting and print submission, **preventing faulty builds from entering production.**
- Conducted destructive and durability testing (tensile loading, mechanical stress, impact testing) on printed shells, characterizing structural failure thresholds.
- Evaluated candidate materials via FEA against strength and durability requirements, **with findings informing production material selection.**
- Optimized a multi-stage post-processing pipeline (printing, depowdering, vapor smoothing), developing a thickness-dependent recipe that improved finish consistency.
- Tested multiple build orientations during nesting, maximizing parts produced per print session.
- Built MATLAB automation for production file preparation, **cutting preprocessing time by ~60%.**
- Inspected 400+ orthoses weekly for surface/dimensional defects, **maintaining quality standards under high production volume.**
- Authored 9 work instructions (~117 pages) and trained 3+ technicians, standardizing procedures across the production team.

PROJECTS

Contact Lens Dry Processing

Bausch + Lomb

- Applied Pugh matrix and functional decomposition, translating requirements into evaluated, feasible concepts.
- Designed and executed a 120+ trial DOE across 8 conditions, **improving mold-release yield ~19%.**

Intravenous Glucose Monitoring System

St. Peter's Hospital Medical Associates

- Built a traceability matrix linking user needs to design inputs and V&V test cases, ensuring full requirement-to-test coverage.
- Reviewed 15+ FDA/IEC/ISO standards and led concept generation, **delivering a multi-phase V&V plan (FMEA).**